

MANOJ ADHIKARI

Data Scientist

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PROFESSIONAL SUMMARY

Data Scientist with 2 years of experience in advanced analytics in insurance and finance industries. Specialized in data science, machine learning, statistical modeling, and data engineering, with proficiency in Python, SQL, and data visualization tools. Demonstrated expertise in streamlining data pipelines, building predictive models, and delivering actionable insights to drive business success. Adept at supporting ad-hoc data analysis projects and collaborating with cross-functional teams to foster data-driven innovation.

SKILLS

- **Languages & Libraries:** Python, Pandas, Numpy, Pytorch, Tensorflow, Keras, Scikit-Learn, Matplotlib, BeautifulSoup, OpenCV
- **Database:** MySQL, PostgreSQL, MongoDB, Elasticsearch, Redis, Amazon Redshift, Snowflake, Google BigQuery, Azure SQL
- **Business Intelligence & Visualization:** PowerBI, Looker, QuickSight, Tableau
- **Cloud & Tools:** GCP: BigQuery, Dataflow, AWS: S3, SageMaker, Lambda, Glue, Azure AI Studio, Document Intelligence, Databricks, Docker, Apache Airflow, Apache Spark, Git, Kubernetes
- **Frameworks:** Flask, Django REST Framework, Scrapy, FastAPI, Streamlit, NLTK, SpaCy
- **Soft Skills:** Leadership, Teamwork, Collaboration, Agile, Presentation, Creative, Problem-Solving, Analytical mindset

EDUCATION

- Master of Science in Computer Science (GPA: 4.0) | East Tennessee State University Aug 2023- May 2025
Relevant Course: ML, NLP, Data Analytics & Visualization, Software Testing, Research Method Johnson City, Tennessee
- Bachelor of science in Computer Science (GPA 3.68) | Tribhuvan University Aug 2016 – May 2020
Relevant Course: Data Mining & Data Warehouse, Analysis of Algorithm, Artificial Intelligence Kathmandu, Nepal

PROFESSIONAL EXPERIENCE

Graduate Teaching Assistant, East Tennessee State University Sep 2023 – Present | Johnson City, Tennessee

- Facilitate Information Technology course (Digital Literacy, Data & AI Literacy, Web Literacy, Cybersecurity etc) and oversee lab projects to 300+ undergraduate students, enhancing their readiness for the digital workforce.
- Mentor new graduate teaching assistant pool on instructional strategies teaching and course resource management.

Data Scientist-II, Renegade Insurance

Feb 2022 - July 2023 | Atlanta, Georgia

- Perform in-depth analysis and pattern mining on leads, quotes, coverage, and policy sales data across 20+ states for personal and commercial lines. Develop targeted group campaigns using predictive models, boosting leads by 7%.
- Integrate heterogenous data source, preprocess, and warehouse data from 15+ insurance carriers in, ensuring seamless data flow and real-time availability for analytics and business intelligence.
- Track key performance indicators (KPIs) like policy uptake, retention rates, claims ratios, and customer lifetime value to assess product success and recommend iterative improvements.
- Develop interactive dashboards and incremental reports to track leads, agent performance metrics, and sales trends, and communicate insights to product teams and executives.
- Innovate data-driven solutions in conjunction with cross functional team, enhancing operational efficiency and business scalability. Provide support for cross-team ad-hoc data analysis projects.

Data Scientist-I, BitsKraft Pvt. Ltd.

Jan 2021 – Jan 2022 | Kathmandu, Nepal

- Develop tools to automate repetitive tasks, such as data preprocessing, profiling and quality check for analytics ready data.
- Utilize statistical and machine learning, data mining techniques to build predictive models for forecasting, financial risk assessment, and customer behavior analysis.
- Prepare and present analytical findings to senior management, stakeholders business units in a clear and concise manner through reports and dashboards.
- Analyze financial performance metrics, risk indicators, and key performance indicators (KPIs).

PROJECTS

- **BucBuddy-University Assistance System:** Advanced RAG architecture based conversational & context aware intelligent Q&A system designed to address complex queries for current & prospective ETSU scholars, featuring advanced accessibility capabilities. *Tools: Langchain, Llama 2, ChromaDB, Hugging Face, Sentence Transformers, Flask. (Funded- [Small Grants in Support](#)).*
- **Phishing Awareness and Simulation System (REST APIs):** Phishing Simulation System with multi-factor authentication with feature of interactive analytical dashboards & reports on breach data, template creation, role, & permission management. *Tools: Python, Django REST framework, SQL.*

- **Network Intrusion Detection System:** Developed predictive model with 92% validation accuracy to correctly identify network breaches through the network data packet analysis using Machine Learning Algorithms & Neural Network models. *Tools: Pytorch, Tensorflow, Keras, Pandas, Numpy, Scikit Learn, Django REST Framework.*

PUBLICATIONS

- **Vision-Based Driver Drowsiness & Distraction Detection Through Behavioral Indicators of Fatigue** (*Author, IEEE SoutheastCon. 2025, Signal & Image Processing, Accepted, Forthcoming*)
 - Led a team of 3 in research and development, formulated research methodology, developed pseudocode. Performed validation and evaluation of proposed methods.
- **Real-Time Driver Drowsiness Detection Using CNN, MediaPipe& ML Classifiers** (*Co-Author, IEE SoutheastCon. 2025, Industrial Applications: Artificial Intelligence, Accepted, Forthcoming*)
 - Preprocessed and augmented datasets, applied transfer learning on VGG16 and ResNet50, and implemented MediaPipe.
 - Performed hyper parameter tuning on traditional classifiers.
 - Performed Principal Component Analysis (PCA) on features and document observations.
- **AIDE: Leveraging Retrieval-Augmented Generation for Context-Aware Educational Data Retrieval and Dialogue** (*Author, IEEE SmartNet 2025, Generative AI in Education, In peer review*)
 - Led a team of 3 in research and development of AIDE’s information retrieval and response generation architecture.
 - Conducted quantitative and qualitative performance evaluation of system. Oversaw team coordination, and project documentation.

PROFESSIONAL DEVELOPMENT

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| <ul style="list-style-type: none">• Foundation of Generative AI• Building RAG Solutions with Azure AI Studio• Data Science Web App: Streamlit and Python• AI Show: Federated Learning with Azure Machine Learning• Creating Multi Task Models With Keras | <ul style="list-style-type: none">• Data Engineering Foundations• Students in Research CITI Program• Machine Learning with Scikit-Learn.• Getting Started with Data Analytics on AWS.• Data Science for Business |
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HONORS AND AWARDS

- The Fr. Martin P. Coyne, S.J Memorial Award for Outstanding Moral Uprightness (2020)
- Fr. Locke and Stiller Research Awardee - Cycle II (2019) | Assessment of Criminal Activities in Kathmandu Valley using KNN Classifier (*Research Annals of Xavier’s Nepal, ISSN:2705-456, Issue: 2.0, 2021*)

EXTRA CURRICULAR ACTIVITIES

Cricketer’s Club of ETSU, <i>Vice President</i>	<i>March 2024 - Present</i>
Renaissance Camp ETSU, <i>Leader, Counselor</i>	<i>June 2024 - August 2024</i>

LANGUAGES

English	<i>Proficient (IELTS Score: 7.0)</i>
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